

CSE226 Final Project Technical Documentation

Multi-Tenant Inter-Office Memo Management System

Deterministic Sequential Approval Workflow Engine & Document Governance SaaS

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Course & Semester:	CSE226 Foundations of Vibe Coding - Summer 2026
Deployed System URL:	https://memo-system-pjbj.vercel.app
Source Code Archive:	https://github.com/ZamanShafin/memo-system/raw/main/source_code.zip
GitHub Repository:	https://github.com/ZamanShafin/memo-system

Table of content

1. System Overview & Core Capabilities.....	3
2. Requirements Implemented (Compliance Matrix)	3
3. Technology Stack	4
4. System Architecture & Diagram	6
5. Database Design & Multi-Tenancy Enforcement.....	7
6. Sequential Workflow Design & Turn Governance	8
7. Security Architecture & Access Control	10
8. Known Limitations	10
9. Deployment Information & Demonstration Accounts	10

1. System Overview & Core Capabilities

The Inter-Office Memo Management System is a full-stack, multi-tenant SaaS application built to digitize, govern, and audit corporate memorandum workflows. It eliminates fragmented email threads and physical paper slips by enforcing deterministic sequential approvals, active delegation, immutable version history, and cryptographic PDF generation.

Key System Highlights

- **Multi-Tenant Isolation:** Logical data partitioning with non-nullable org_id scoping across every database query.
- **Strict Turn Governance:** Memos execute sequentially (Requester -> Dept Head -> Finance -> Director -> CEO). Later participants are physically blocked from acting early.
- **Active Delegation:** Temporary date-bounded approval authority handoffs when officers are on leave.
- **Dynamic Reviewer Injection:** Approvers can dynamically insert specialized reviewers (e.g. Legal Counsel) mid-stream.
- **Tamper-Evident Auditing:** Append-only audit logs with timestamps, actor IDs, comments, and client IPs.
- **PDF Seal Engine:** Generates official corporate PDFs complete with digital signatures and QR verification codes.

2. Requirements Implemented (Compliance Matrix)

Ref #	Requirement Area	Implementation Highlights
Req 1-4	Multi-Tenancy & RBAC	Complete org_id data perimeters, Admin/User roles, department management.
Req 5-7	Auth & Memo Authoring	JWT tokens, password reset, Quill.js rich text editor, priority tags, file uploads.

Req 8-10	Sequential Workflow Engine	Strict linear turn execution, workflow templates, mid-stream reviewer insertion.
Req 11-13	Decisions & Delegation	Approve, Reject, Request Changes with audit notes; date-bounded delegation.
Req 14-17	PDFs, Alerts & Versions	Cryptographic PDF seal generation, notification tray, immutable version snapshots.
Req 18-22	Audit & Security	Append-only audit logs, turnaround charts, admin self-deactivation protection.
Req 23-30	Deployment & Testing	Live on Vercel, sanitized source ZIP, 20/20 passing pytest suite, full prompt log.

3. Technology Stack

Layer / Component	Technology & Libraries
Backend API Framework	FastAPI (Asynchronous Python 3.10+ REST framework)
Database Engine & ORM	PostgreSQL 16 (Neon Serverless Cloud DB) + SQLAlchemy 2.0 ORM

Authentication & Security	PyJWT (JSON Web Tokens) + Passlib (PBKDF2-SHA256 & Bcrypt hashing)
Frontend Single Page App	Vanilla JavaScript SPA engine (app.js) with zero build-step overhead
Styling & Design System	Tailwind CSS utility framework + Lucide Icons + Google Fonts
Rich Text Authoring	Quill.js WYSIWYG editor with custom toolbar & HTML sanitization
Visualizations & Charts	Chart.js for real-time status doughnuts and turnaround duration bars
Document / PDF Engine	ReportLab & xhtml2pdf with embedded signature stamps and QR codes
Cloud Infrastructure	Vercel Serverless Functions with automated CI/CD pipeline
Testing & Tooling	Pytest automated test harness (20/20 tests passing) + Astral uv

4. System Architecture & Diagram

The application is structured into a clean 3-tier decoupled architecture comprising a Client SPA, a Stateless FastAPI REST API service layer, and a cloud-native PostgreSQL relational database.



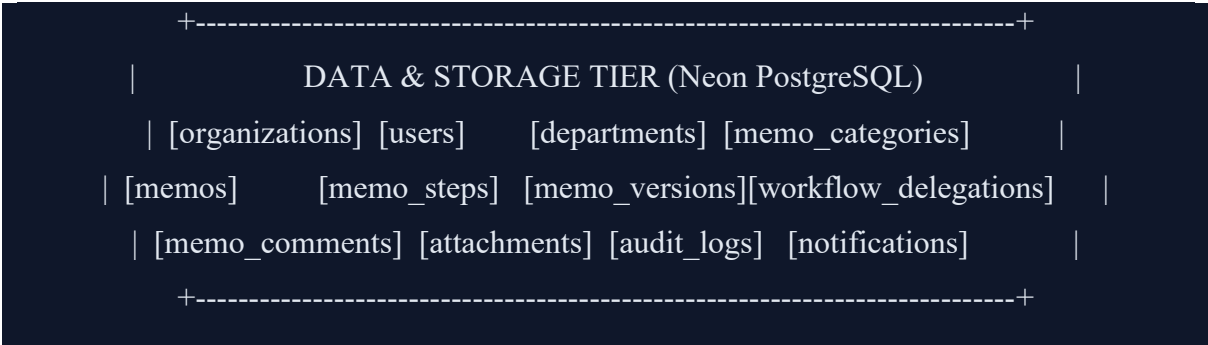


Figure: High-Level 3-Tier System Architecture Diagram

5. Database Design & Multi-Tenancy Enforcement

Multi-tenancy is enforced through strict logical data partitioning. Every single table in the database contains a non-nullable foreign key referencing organizations.id. Multi-tenancy is guaranteed at the database and API layer through the following mechanisms:

- Server-Side Query Scoping: Every database query in all routers automatically includes `models.Entity.org_id == current_user.org_id`. Client-side filtering is never relied upon for security.
- Direct Object Access Validation: When accessing a resource by ID (e.g. GET /memos/10), the server verifies `memo.org_id == current_user.org_id`. If the tenant IDs mismatch, an immediate HTTP 404/403 is thrown.
- Cascade Isolation: Deleting or modifying a department, user, or workflow step is strictly constrained to the user's organization.

ORGANIZATION		USER		MEMO	
1 *		1 *			
id (PK)		id (PK)		id (PK)	
name, code		org_id (FK)		org_id (FK)	
contact_email		department_id(FK)		memo_number, title	
		full_name, email		body_html, priority	
		role, is_active		status, author_id	
				current_assignee_id	

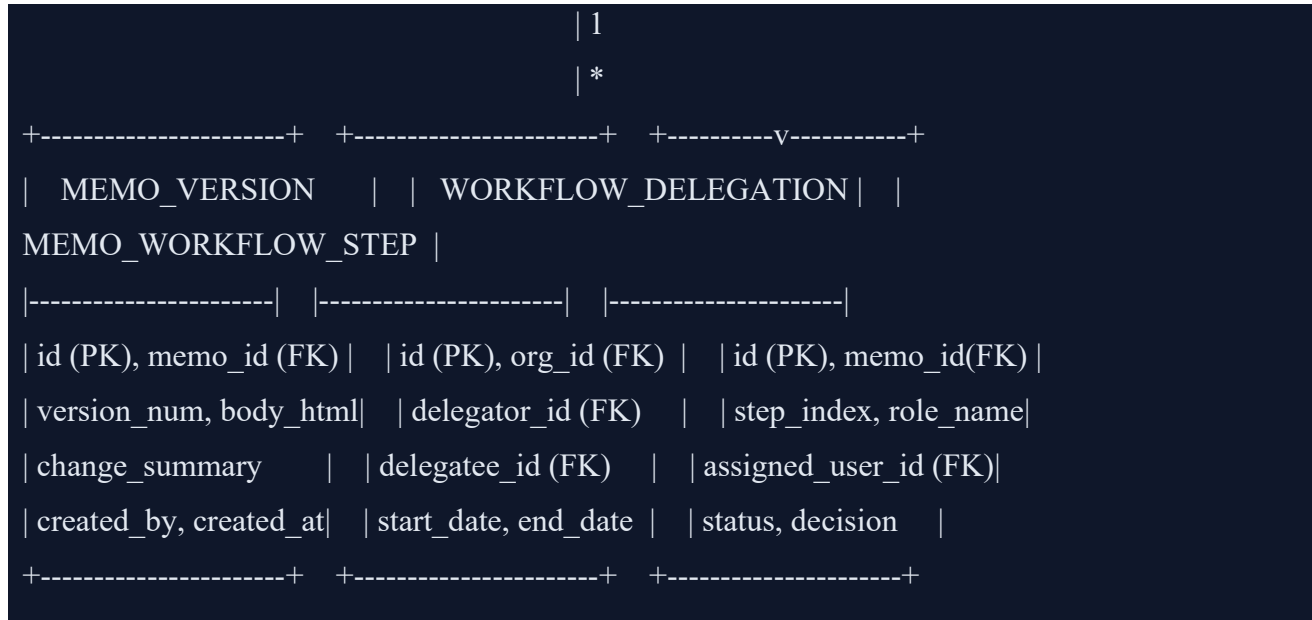
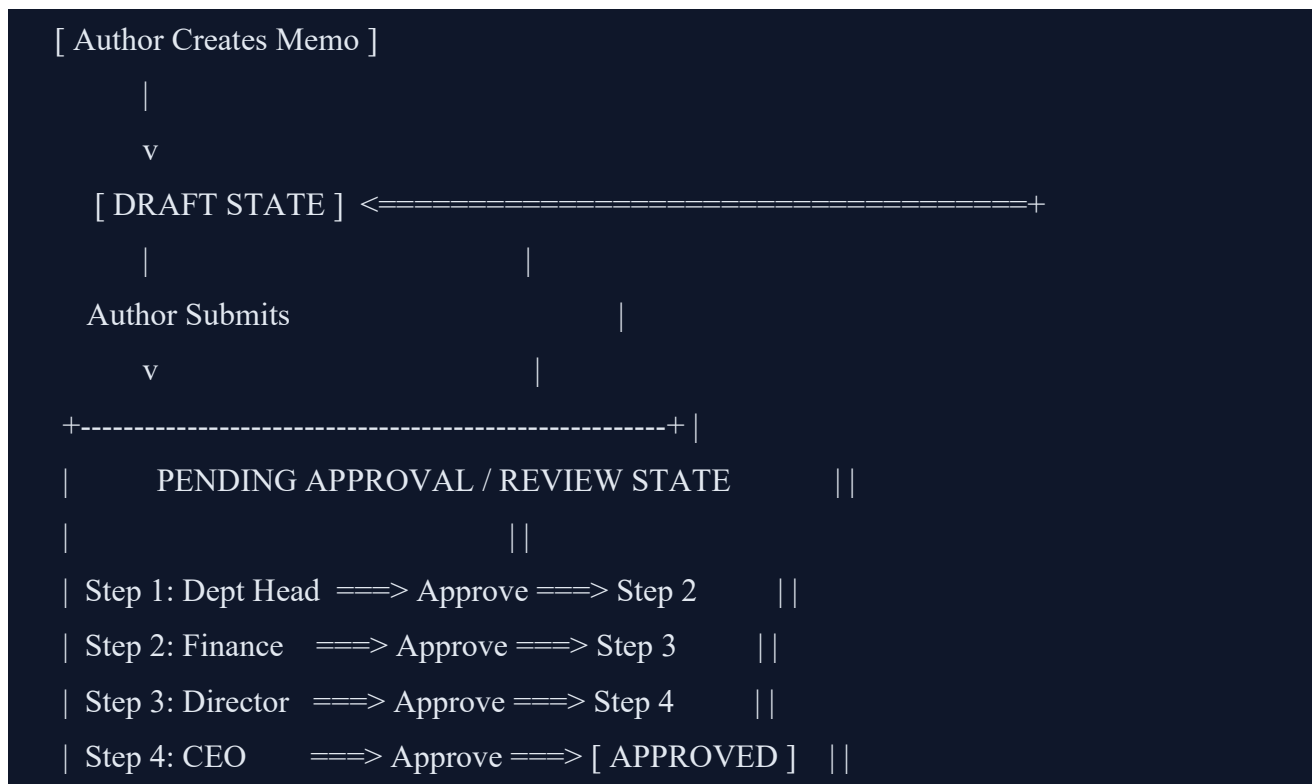


Figure: Entity-Relationship & Multi-Tenancy Diagram

6. Sequential Workflow Design & Turn Governance

The sequential workflow engine operates as a deterministic finite-state machine with strict turn-based locks. Only the currently assigned participant (or their authorized active delegate) can sign off.



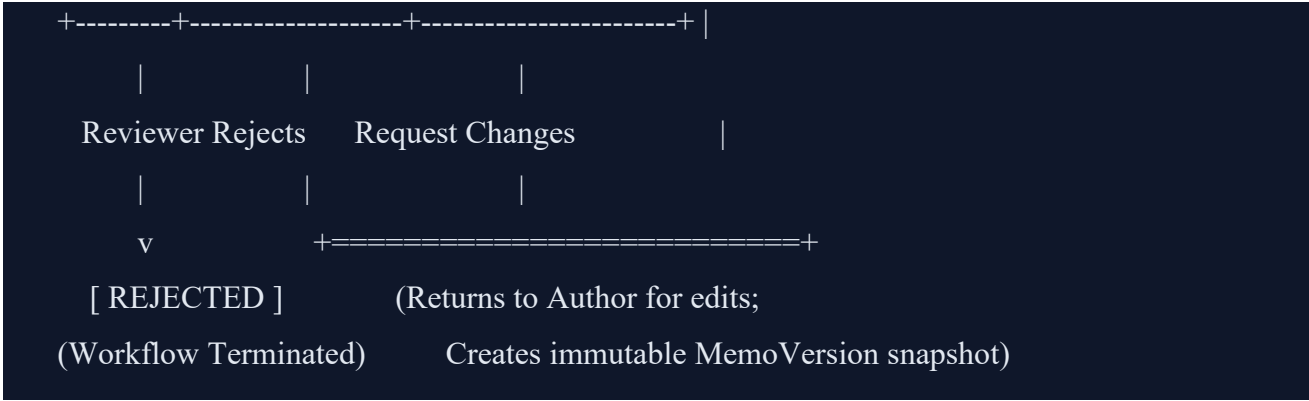


Figure: Sequential Approval Workflow State Machine Diagram

Workflow Stage / Rule	Operational Behavior & Governance
Draft -> Submission	Author drafts memo and defines sequence. Submitting advances turn to Step 1 and dispatches notifications.
Step-by-Step Turns	Step N must be completed before Step N+1 can act. Later approvers are physically blocked by server-side checks.
Dynamic Reviewer Insertion	Approver can select 'Approve & Add Reviewer' to insert a specialist (e.g. Legal Counsel) before next step.
Decline & Reroute	Turn holder can reassign their active step to an eligible colleague in the same organization.
Delegation Protocol	When active date window is valid, designated delegatee can sign off on behalf of delegator with full audit attribution.
Changes Requested Cycle	Reviewer returns memo with comments. Author

	updates content; system creates MemoVersion snapshot before resubmission.
Final Approval & Seal	Final approver (CEO) marks workflow Approved. System locks content as read-only and generate official signed PDF.

7. Security Architecture & Access Control

- Cryptographic Passwords: Passwords are salted and encrypted using PBKDF2-SHA256 and Bcrypt algorithms.
- JWT Bearer Sessions: Stateless tokens signed with HMAC-SHA256 with 24-hour expiration.
- Admin Self-Protection: Dedicated server-side rules block administrators from deactivating or demoting their own accounts.
- Injection & XSS Defenses: Parameterized queries via SQLAlchemy prevent SQL injection; rich text HTML is sanitized before rendering.
- Secure Cloud Connection: Database connection strictly requires SSL/TLS encryption (sslmode=require).

8. Known Limitations

1. In-App Notification Transport: Uses responsive client-side HTTP polling rather than full-duplex WebSockets.
2. Object Storage: File attachments are persisted within the PostgreSQL database layer rather than external AWS S3 buckets.
3. Two-Factor Authentication: MFA/TOTP is reserved for future enterprise roadmap versions.

9. Deployment Information & Demonstration Accounts

- Live Deployed URL: <https://memo-system-pjbj.vercel.app>
- Source Code ZIP: https://github.com/ZamanShafin/memo-system/raw/main/source_code.zip
- Active Organization Code: acme
- Universal Password: password123

Role	Login Email	Persona / Purpose
Admin	admin@acmecorp.com	Sarah Jenkins (System Administrator)
Author	alex.morgan@acmecorp.com	Alex Morgan (Senior Engineer / Author)
Dept Head	head.eng@acmecorp.com	David Vance (VP of Engineering / Approver)
Delegate	jessica.taylor@acmecorp.com	Jessica Taylor (Acting Delegate for David)
Finance	finance.mgr@acmecorp.com	Rachel Green (Chief Financial Manager)
Director	director@acmecorp.com	Marcus Sterling (Director of Operations)
CEO	ceo@acmecorp.com	Eleanor Vance (Chief Executive Officer)